

Competition and Inflation under HSA Demand

Nine-Cell State-Space Implementation Report

LONG CORE RUN – MANDATORY ROBUSTNESS INCOMPLETE

Abstract

This report is generated by the executable implementation of the design in `design.tex`. It estimates the pre-declared 3×3 price–activity matrix using a measurement-only business-demography cut, paired quarterly annualized and exactly aggregated year-over-year likelihoods, and a secondary full-joint information-flow sensitivity. The displayed run is explicitly labeled above. A test run validates data alignment, samplers, aggregation, outputs, and rendering; it is not an empirical result and does not pass production convergence requirements.

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1 Status and interpretation boundary

The primary coefficients are conditional varying-coefficient associations. The code does not relabel observed firm counts as theoretical effective varieties, does not equate aggregate inverse markup with structural marginal cost, and does not treat full-joint feedback as an endogeneity correction. Missing benchmark-admissibility conditions remain visible as failed or conditional gates.

2 Data and executable design

The common sample contains 124 quarterly endpoints from 1982Q1 through 2012Q4. Inflation is $400\Delta \log P_t$ for PPI, headline CPI, and core CPI. The three activity measures are the log inverse-markup proxy, the frozen BN output gap, and the negative unemployment gap. Annual firm counts enter at Q4 only. Quarterly QCEW establishments provide the business-demography indicator. All nine cells share one complete-case sample; cell-specific dropping is forbidden by the loader.

The primary state posterior is the modular cut

$$p_{cut}(S, \beta_c) = p(S \mid C)p(\beta_c \mid \pi_c, X_c, S).$$

The full-joint E2 result is secondary. Its slow and fast state paths are sampled with the two conditional FFBS blocks implied by the bilinear term. For YoY, the entire quarterly conditional mean is premultiplied by A_4 and the likelihood uses $G_4 = A_4 A_4'$.

3 Nine-cell primary coefficients

Table 1: Nine-cell QoQ modular-cut E2 coefficients

Cell	Inflation	Activity	ψ	κ_0	κ_1	θ_0	γ
1	PPI	Inverse markup	0.019 [-1.305, 1.345]	-6.411 [-33.502, 21.314]	1.314 [-5.337, 8.263]	0.019 [-0.792, 0.816]	-0.016 [-0.344, 0.319]
2	PPI	BN output gap	-0.520 [-1.447, 0.535]	0.999 [-0.254, 2.237]	-0.246 [-0.702, 0.245]	0.108 [-0.707, 0.924]	-0.018 [-0.340, 0.311]
3	PPI	Negative unemployment gap	-0.276 [-1.261, 0.736]	0.100 [-0.816, 1.006]	0.072 [-0.333, 0.491]	0.079 [-0.765, 0.894]	0.003 [-0.334, 0.349]
4	CPI	Inverse markup	0.081 [-1.144, 1.298]	1.188 [-13.372, 16.173]	0.596 [-3.819, 4.850]	0.026 [-0.620, 0.694]	-0.075 [-0.323, 0.185]
5	CPI	BN output gap	-0.220 [-0.810, 0.338]	0.439 [-0.046, 0.920]	-0.117 [-0.373, 0.072]	0.153 [-0.521, 0.858]	-0.098 [-0.348, 0.184]
6	CPI	Negative unemployment gap	-0.087 [-0.617, 0.436]	0.199 [-0.184, 0.600]	0.088 [-0.106, 0.308]	0.112 [-0.529, 0.744]	-0.014 [-0.277, 0.253]
7	Core CPI	Inverse markup	0.330 [-0.701, 1.312]	1.479 [-6.751, 10.068]	0.402 [-3.007, 3.567]	0.131 [-0.353, 0.598]	-0.058 [-0.251, 0.157]
8	Core CPI	BN output gap	0.201 [-0.158, 0.605]	0.009 [-0.185, 0.196]	0.017 [-0.097, 0.114]	0.099 [-0.364, 0.562]	-0.049 [-0.227, 0.164]
9	Core CPI	Negative unemployment gap	0.273 [-0.185, 0.779]	0.061 [-0.204, 0.315]	0.071 [-0.078, 0.202]	0.122 [-0.331, 0.559]	0.008 [-0.184, 0.214]

Notes: posterior mean and central 95% credible interval. The test run is a software validation only; its intervals are not inferential results.

4 Identification before interpretation

Gate / evidence	Test-run status	Interpretation
Quarterly measurement information $R_q \leq 0.80$	1.086	Threshold failed; no stationary-state interpretation.
Coefficient learning and convergence	Not evaluated	Short chains intentionally fail the production ESS/ $\hat{R}$ requirements.
Cut-joint information flow	Displayed	Conflicts are diagnostic flags, not efficiency gains to promote.
Fast-by-fast CS1/CS2	Executed	Test-run sensitivity only. Activity endogeneity is not solved.
Persistent-error / low-frequency nuisance	Not run	Reserved for production; report does not imply robustness.
Chib Bayes factors	Not run	No Bayes-factor claim is made from a modular cut.
Real-time LFO score	Inadmissible	Vintage/release-date inputs are incomplete; no forecast claim is made.

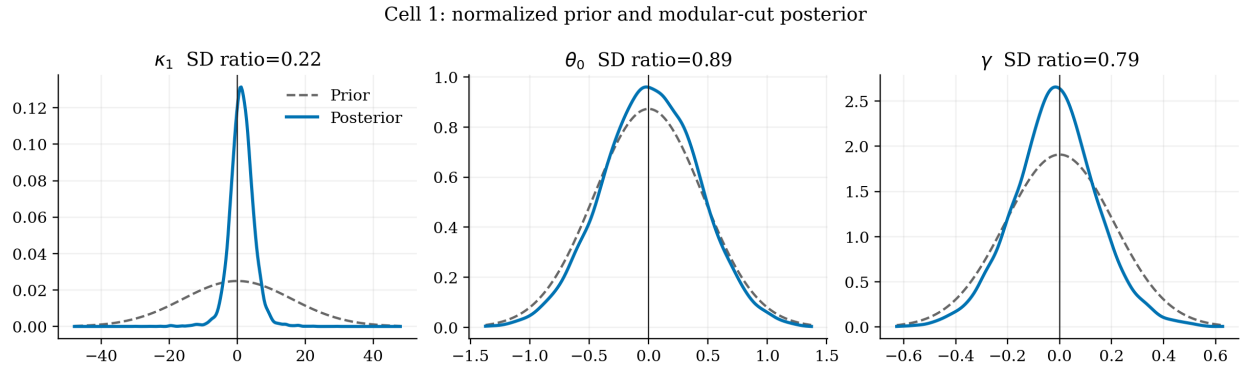


Figure 1: Cell 1 prior and primary modular-cut posterior. Short test chains are shown only to verify the reporting path.

5 Competition-state measurement

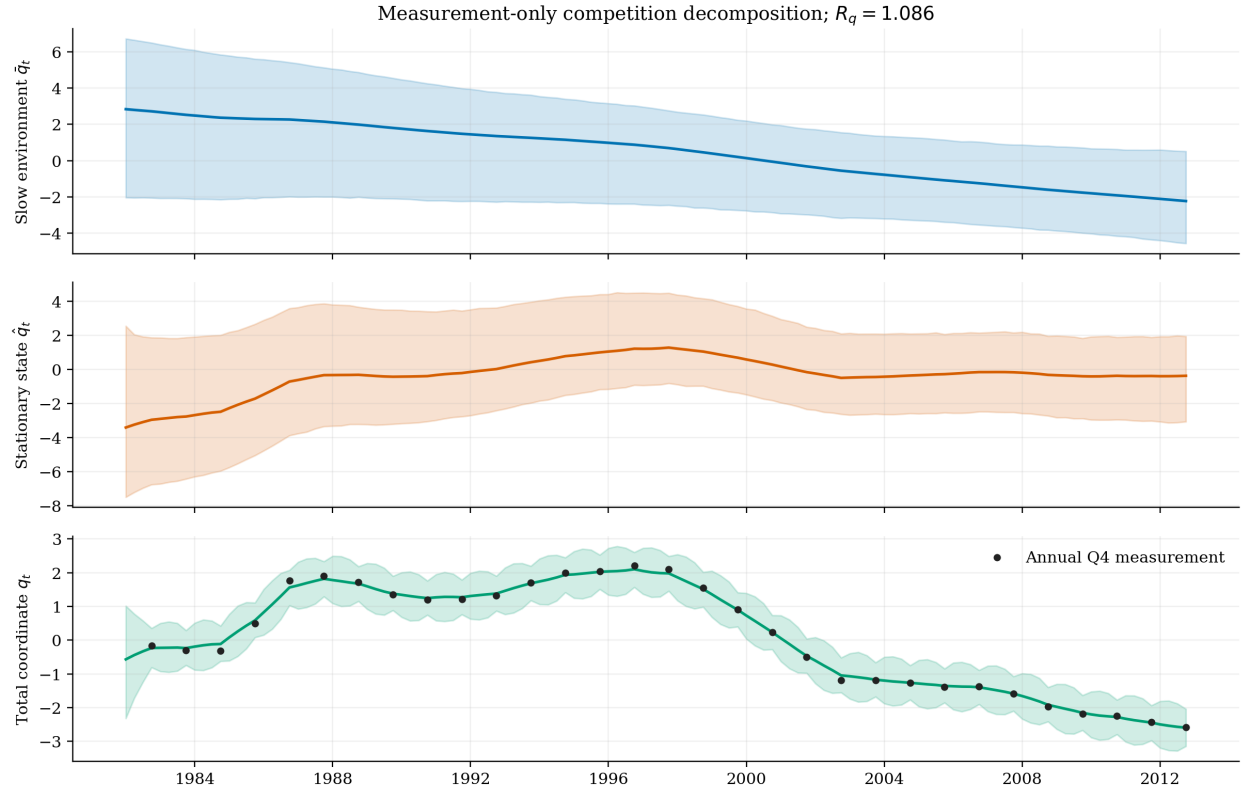


Figure 2: Measurement-only slow, stationary, and total business-demography coordinate.

6 Time-varying slope

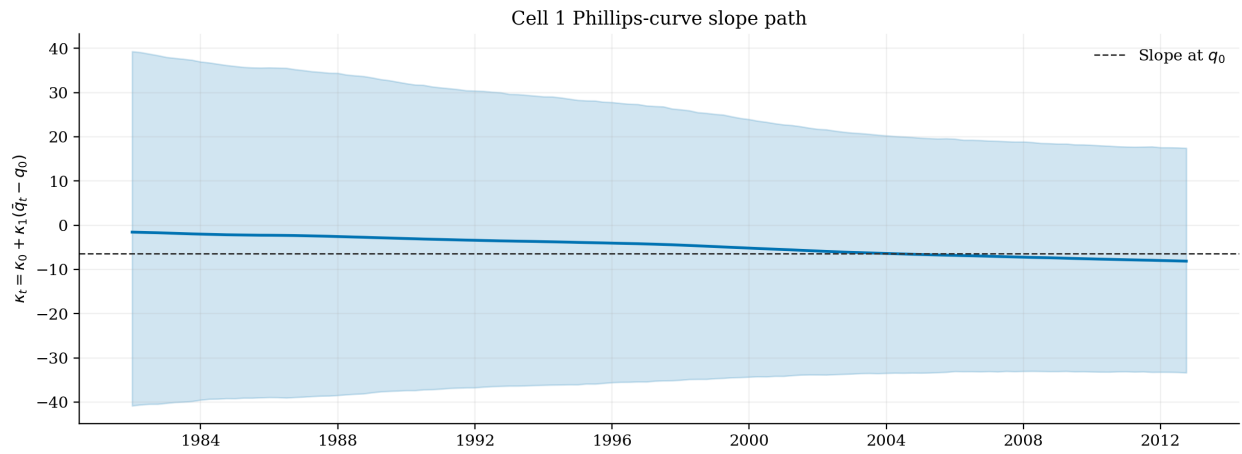


Figure 3: Cell 1 Phillips-curve slope path under the primary cut E2 model.

7 Cut versus full joint

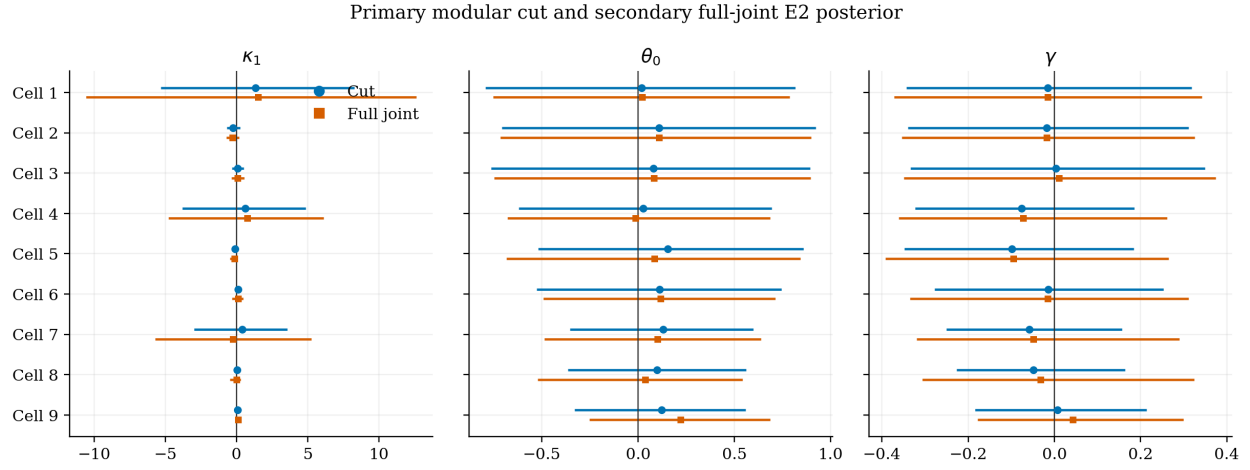


Figure 4: Primary cut and secondary full-joint posterior summaries.

Cell	Maximum shift (cut SD)	Conflicts	Status
1	0.06	0	Pass
2	0.13	0	Pass
3	0.06	0	Pass
4	0.12	1	Flag
5	0.35	0	Pass
6	0.16	0	Pass
7	0.40	1	Flag
8	0.58	1	Flag
9	0.44	0	Pass

8 Paired QoQ and exact YoY transformation

Table 2: Endpoint-matched QoQ and exactly aggregated YoY comparison

Cell	Inflation	Activity	κ_1 QoQ / YoY	θ_0 QoQ / YoY	γ QoQ / YoY
1	PPI	Inverse markup	1.231 [-5.566, 8.152] / 1.493 [-5.157, 8.366]	0.017 [-0.797, 0.808] / 0.006 [-0.797, 0.824]	-0.018 [-0.353, 0.321] / -0.024 [-0.362, 0.317]
2	PPI	BN output gap	-0.228 [-0.697, 0.278] / -0.239 [-0.677, 0.240]	0.105 [-0.705, 0.909] / 0.100 [-0.713, 0.917]	-0.026 [-0.355, 0.313] / -0.029 [-0.345, 0.305]
3	PPI	Negative unemployment gap	0.078 [-0.329, 0.494] / 0.082 [-0.322, 0.489]	0.072 [-0.734, 0.902] / 0.065 [-0.761, 0.872]	-0.001 [-0.352, 0.338] / -0.004 [-0.338, 0.325]
4	CPI	Inverse markup	0.390 [-3.992, 4.804] / 0.652 [-3.743, 4.980]	0.041 [-0.636, 0.745] / 0.021 [-0.619, 0.690]	-0.088 [-0.350, 0.207] / -0.084 [-0.337, 0.192]
5	CPI	BN output gap	-0.091 [-0.370, 0.094] / -0.113 [-0.365, 0.071]	0.174 [-0.540, 0.912] / 0.151 [-0.534, 0.849]	-0.124 [-0.388, 0.187] / -0.106 [-0.359, 0.182]
6	CPI	Negative unemployment gap	0.111 [-0.106, 0.346] / 0.088 [-0.110, 0.311]	0.137 [-0.525, 0.817] / 0.105 [-0.551, 0.743]	-0.021 [-0.293, 0.247] / -0.022 [-0.284, 0.256]
7	Core CPI	Inverse markup	-0.206 [-4.042, 3.252] / 0.426 [-3.078, 3.597]	0.184 [-0.374, 0.705] / 0.131 [-0.357, 0.596]	-0.086 [-0.296, 0.158] / -0.060 [-0.254, 0.160]
8	Core CPI	BN output gap	0.051 [-0.106, 0.189] / 0.018 [-0.094, 0.119]	0.185 [-0.396, 0.721] / 0.101 [-0.356, 0.559]	-0.107 [-0.315, 0.152] / -0.054 [-0.240, 0.158]
9	Core CPI	Negative unemployment gap	0.091 [-0.092, 0.247] / 0.073 [-0.074, 0.204]	0.191 [-0.321, 0.666] / 0.121 [-0.341, 0.572]	-0.013 [-0.209, 0.215] / 0.004 [-0.182, 0.211]

Notes: QoQ uses the YoY endpoint dates. YoY aggregates the full quarterly design and uses $G_4 = A_4 A_4'$. A posterior for the QoQ-minus-YoY difference is deliberately not reported.

9 Economic magnitudes

Cell	Inflation	Activity	IQR-normalized slope contribution (pp)
1	PPI	Inverse markup	0.150 [-0.607, 0.940]
2	PPI	BN output gap	-1.146 [-3.271, 1.141]
3	PPI	Negative unemployment gap	0.397 [-1.845, 2.720]
4	CPI	Inverse markup	0.068 [-0.435, 0.552]
5	CPI	BN output gap	-0.543 [-1.737, 0.335]
6	CPI	Negative unemployment gap	0.490 [-0.588, 1.706]
7	Core CPI	Inverse markup	0.046 [-0.342, 0.406]
8	Core CPI	BN output gap	0.077 [-0.451, 0.532]
9	Core CPI	Negative unemployment gap	0.395 [-0.430, 1.121]

The table reports an IQR-by-IQR slope-change contribution in annualized inflation percentage points. It enables comparison in common outcome units without treating raw coefficients on differently scaled activity variables as comparable structural objects.

10 Auxiliary HSA benchmark

Item	Result
Status	conditional benchmark
Local QoQ endpoints	45
ζ_{ref} / b_x	6.00 / 1.00
δ_{HSA}	0.118 [-14.210, 14.567]
Equivalence band	± 1.599
Equivalence probability	0.171

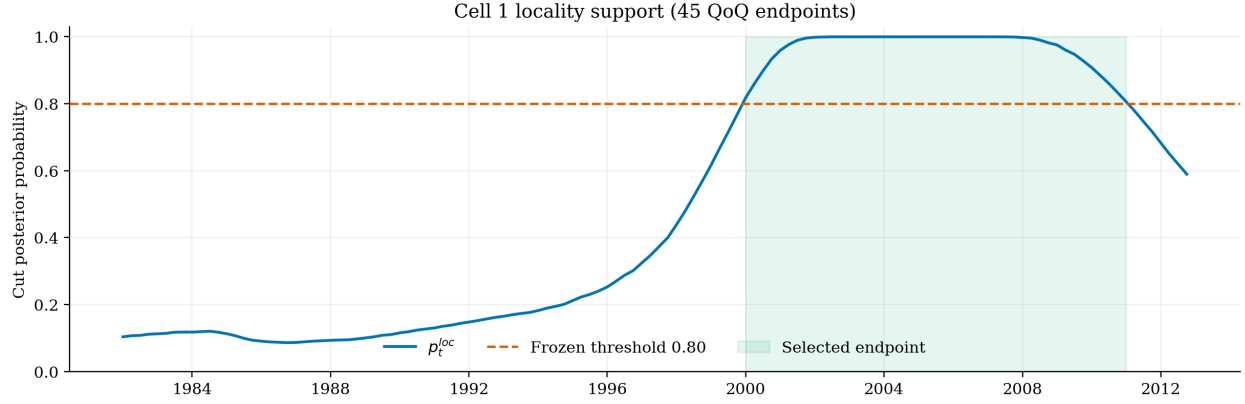


Figure 5: Frozen local-support rule based only on the measurement cut.

The benchmark is conditional on the unit-scale $b_x = 1$ calibration, maintained activity orthogonality, a GDP-deflator expectation proxy for PPI, and a calibrated reference markup. Compatibility is not a probability that HSA is true.

11 What this run does not establish

This run does not establish causal competition effects, a structural activity equation, real-time predictive validity, or a fully admissible PPI benchmark. Persistent-error, low-frequency-nuisance, alternative-state-law, prior-scale, Chib, and real-time LFO exercises are not silently treated as passed when they were not run. Production interpretation requires all pre-declared simulations, robustness exercises, and convergence gates in `design.tex`.